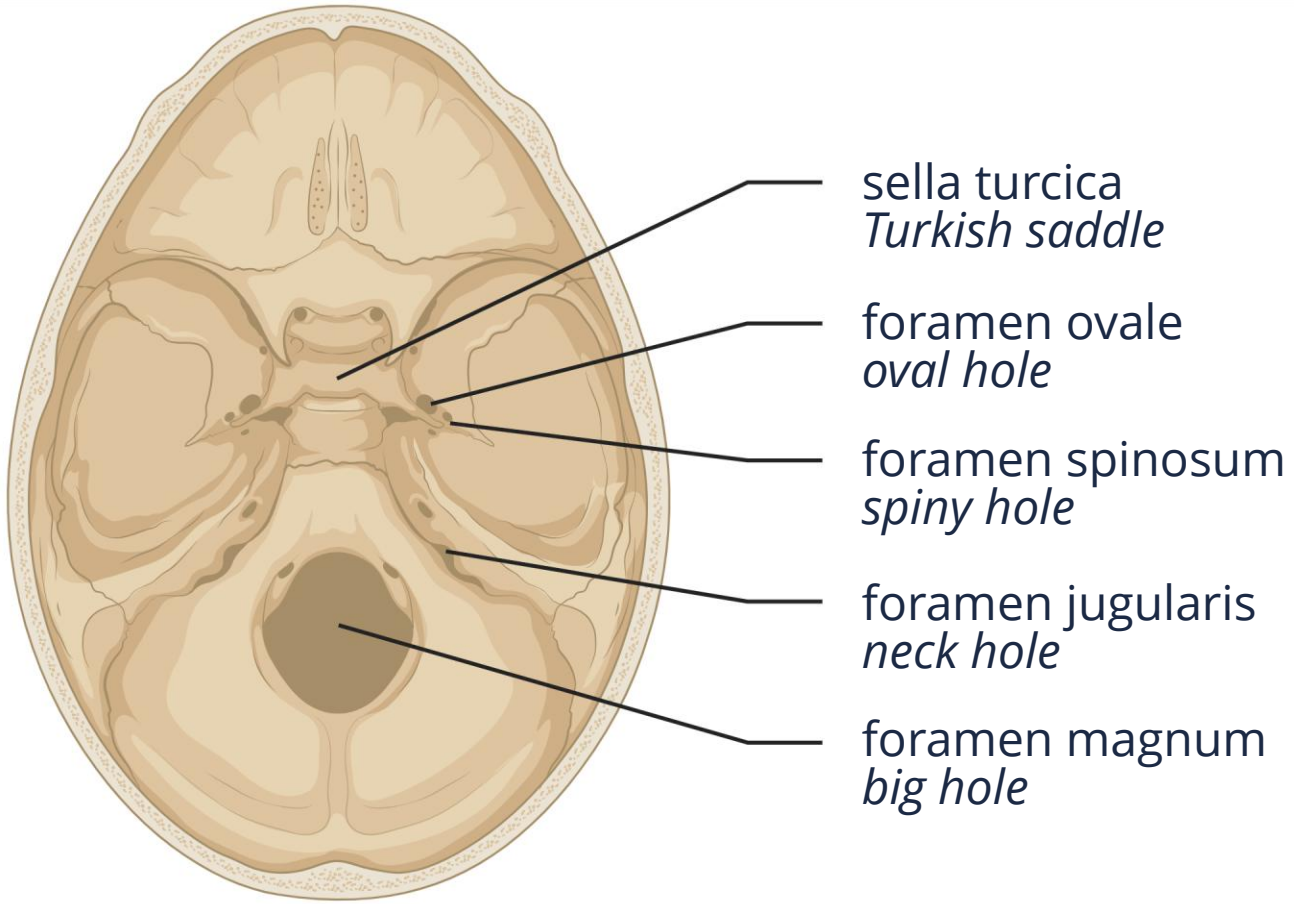
It is easier to learn patterns if you know the language



Foundation models can learn general sequence ‘language’ from a large corpus of text

Learn the language from a general task
Predict the missing word

A sentence _____ words.

A sentence eats words. ✗

A sentence contains words. ✓

Predict the missing nucleotides

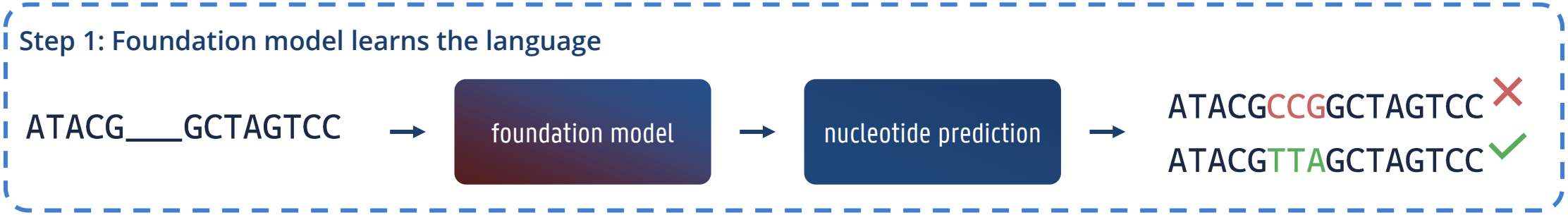
ATACG___GCTAGTCC

ATACGCCGGCTAGTCC ✗

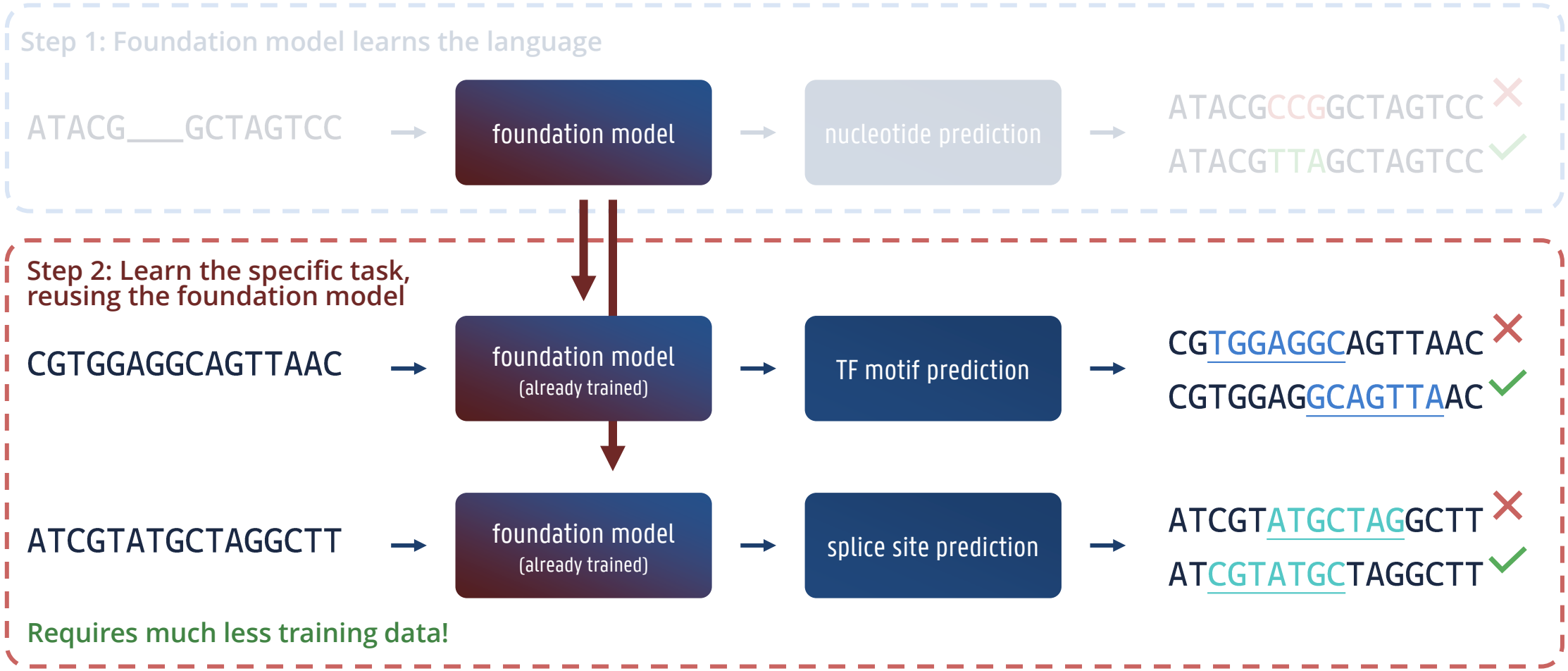
ATACGTTAGCTAGTCC ✓

No exact labels required
Usually, a lot of data available

Pretraining & fine-tuning turns one model into many predictors

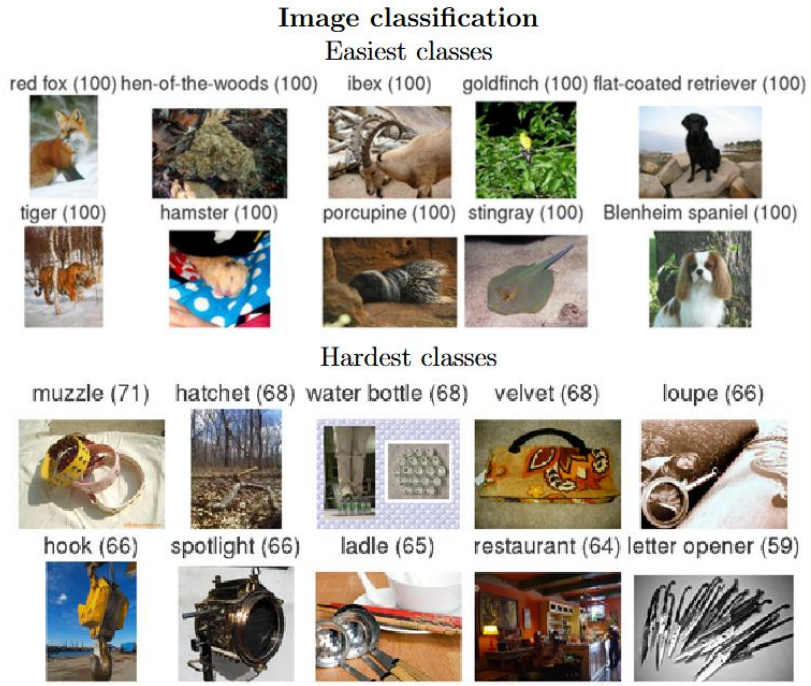


Pretraining & fine-tuning turns one model into many predictors

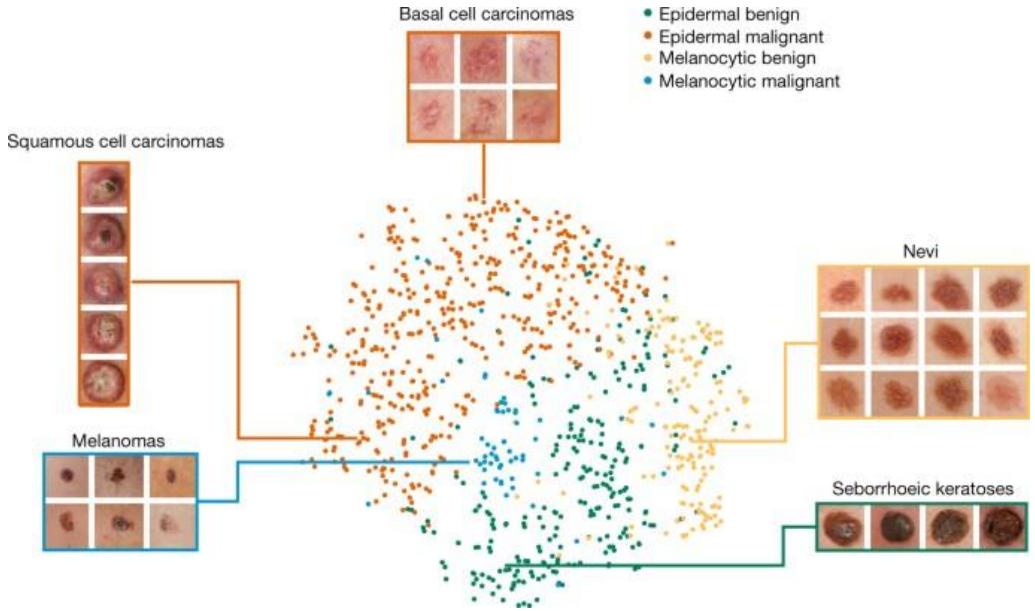


Foundation models can be highly generic, while the subsequent finetuning task can be very domain-specific

Inception v3, as foundation model
trained on >1m images of 1k object categories



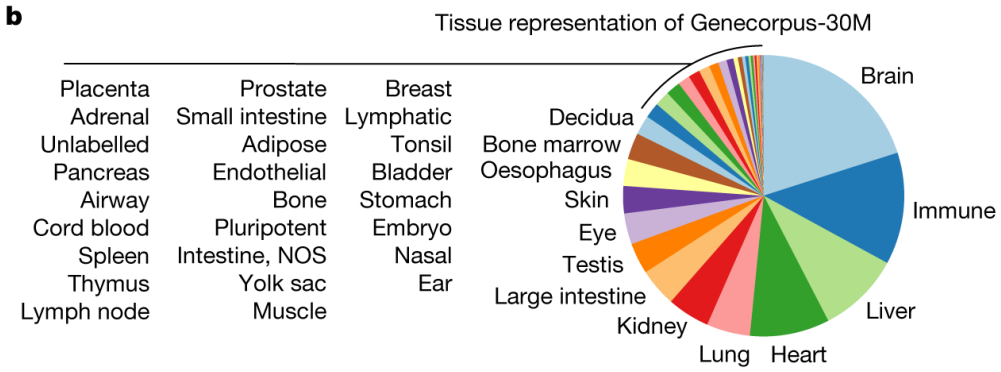
Finetuned for skin lesion classification
finetuned with 127,463 images of skin lesions



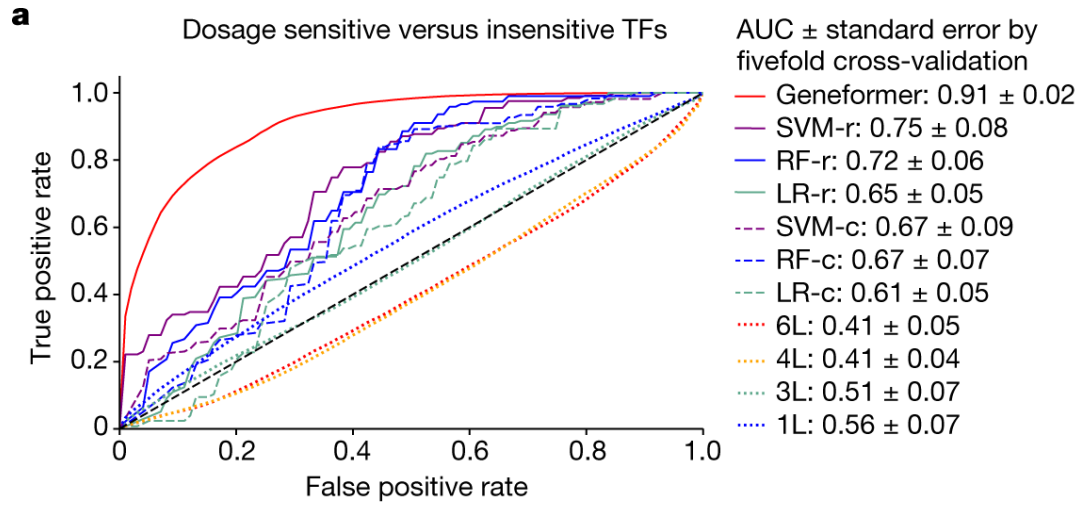
Esteva, A. et al. Dermatologist-level classification of skin cancer with deep neural networks. Nature 542, 115–118 (2017). doi:10.1038/nature21056

Foundation models can also be more domain-specific, but exploit large corpuses of unlabeled data

Pretraining on transcriptomics data
~30 million single cell transcriptome samples



Finetuned for copy number variation gene dosage sensitivity
10.000 transcription factors with known sensitivity



Amino acids can be directly encoded in various ways

Manual feature engineering

ACDTK



	A	C	D	T	K
Basi	37	35	59	129	191
Heli	68	23	33	29	32
Hydr	51	75	25	35	0
pI	32	23	0	27	69

one-hot encoding

ACDTK



	A	C	D	T	K
A?	1	0	0	0	0
C?	0	1	0	0	0
D?	0	0	1	0	0
…	…	…	…	…	…
T?	0	0	0	1	0
…	…	…	…	…	…
K?	0	0	0	0	1
…	…	…	…	…	…

learned embedding

ACDTK



A	C	D	T	K
0.73	0.14	0.88	0.46	0.21
0.57	0.95	0.32	0.69	0.04
0.81	0.27	0.53	0.10	0.76
0.38	0.62	0.17	0.84	0.49
0.92	0.35	0.58	0.07	0.64
0.11	0.79	0.43	0.96	0.28
0.67	0.24	0.85	0.51	0.13
0.40	0.73	0.19	0.60	0.88

In summary, pretraining and finetuning serve two different goals

	Direct training	**Pretraining**	**Finetuning**
Objective	*Learn the 'language' & the specific task at once*	Learn the general 'language' of the training task	Learn the specific training task
Method	*Supervised*	Supervised (with labels) or self-supervised (masking)	Supervised (with labels)
Data requirement	*Large corpus of examples*	Very large corpus of examples	Much smaller data set*
Downstream applicability	*Very specific*	Very diverse	Very specific

* depending on how different the downstream task is from the foundation model.

Recap

- Foundation models first learn the general structure of a data domain from large datasets.
- This pretraining step is often self-supervised, so exact labels are not always required.
- The pretrained model can then be fine-tuned for specific downstream tasks.
- This is very useful in biomedicine, where labeled data are often scarce but unlabeled data can be abundant.